

Dr. Lakshmana dora Chandrala

Assistant Professor, Department of Mechanical and Aerospace Engineering, IIT Hyderabad

Email: lchandrala@mae.iith.ac.in, Phone: +91 8179069729

RESEARCH AREAS

- Compressible flows, Optical Holography, Multiphase flows, Machine learning.

EDUCATION

- PhD in Aerospace Engineering, IIT Kanpur, India (2016), CGPA: 10/10
- M.Tech in Aerospace Engineering, IIT Kanpur, India (2009), CGPA: 9/10
- B. E., Mechanical Engineering, Andhra University, India (2007), First class with distinction, 80/100

EMPLOYMENT HISTORY

- Assistant Professor, Department of Aerospace and Mechanical Engineering, IIT Hyderabad, India (July 2020 - Present)
- Post-doctoral fellow, Johns Hopkins University, Baltimore, USA (2016 - 2020)

JOURNAL PUBLICATIONS

[Google Scholar link](#)

Publications after joining IITH

1. L.M. Sharma, H.N. Dixit, L.D. Chandrala*, (2025). Scaling and dynamics of buoyant immiscible liquid jets in the laminar regime, **Physical Review Fluids**, 10(10), 104303.
2. L.M. Sharma, H.N. Dixit, L.D. Chandrala*, (2025). Capillary pinch-off in liquid-in-liquid systems, **The European Physical Journal Special Topics**, 1-10.
3. V. K. Warghat, Y. Biswal, S. Sharma, P. S. Kolhe, L. D. Chandrala*, K. C. Sahu., (2025). Jet breakup dynamics of viscoelastic carboxymethyl cellulose solutions, **International Journal of Multiphase Flow**, 192, 10534.
4. Vinod, V., Majji, G. P., Shinde. V A., Ram, O., and Chandrala L.D.*, (2025). Attenuation of blast waves using rigid porous plate arrays: Insights from experiments and simulations, **Physics of Fluids**, 37(7), 1070-6631.
5. Chakraborty, S., Ade, S. S., Devsoth, L., Chandrala, L. D., Prabhakaran, T., Matar, O. K., and Sahu, K. C. (2025). Drop size distribution from laboratory experiments based on single-drop fragmentation and comparison with aerial in-situ measurements. **Journal of the Atmospheric Sciences**, 92 (8), 1441-1450
6. Chakraborty, S., Ade, S. S., Chandrala, L. D., and Sahu, K. C. (2025). Droplet breakup morphologies and the resultant size distribution in an opposed-flow airstream at different Weber numbers. **International Journal of Multiphase Flow**, 191, 105310.
7. Gupta, C., Choudhury, A., Chandrala, L. D., and Dixit, H. N. (2024). An experimental study of flow near an advancing contact line: a rigorous test of theoretical models. **Journal of Fluid Mechanics**, 1000, A45.
8. Vinod, V., Majji, G. P., Vootla, S., Ram, O., and Chandrala, L. D*. (2024). Predicting pressure buildup behind perforated plates under blast wave impact: A simplified approach. **Physics of Fluids**, 36(11), 116111.
9. Gupta, C., Chandrala, L. D., and Dixit, H. N. (2024). An experimental investigation of flow fields near a liquid-liquid moving contact line. **The European Physical Journal Special Topics**, 233(8), 1653-1663.
10. Ade, S. S., Kirar, P. K., Chandrala, L. D., and Sahu, K. C. (2024). Droplet breakup and size distribution in an airstream: Effect of inertia. **Physical Review Fluids**, 9, 084004.
11. Vinod V., Rao Divya, Chandrala, L. D*. (2024). A Study of Shock-induced Transient Jets: the Impact of Exit Conditions on Mach Disk and Vortex Ring. **Journal of Fluids Engineering**. 146 (8).
12. Ade, S. S., Gupta, D., Chandrala, L. D., and Sahu, K. C. (2024). Application of deep learning and inline holography to estimate the droplet size distribution. **International Journal of Multiphase Flow**, 177, 104853.
13. Ade, S. S., Chandrala, L. D.*, and Sahu, K. C. (2023). Size distribution of a drop undergoing breakup at moderate Weber numbers. **Journal of Fluid Mechanics**. 959, A38.
14. Ade, S. S., Kirar, P. K., Chandrala, L. D. *, and Sahu, K. C. (2023). Droplet size distribution in a swirl airstream using in-line holography technique. **Journal of Fluid Mechanics**, 954, A39.
15. Baishakhi Ghosh, Nishida K, Chandrala, L.D, Saborny Mahmud, Shreeti Thapa, Carter Swaby, Si Chen, Atulya Aman Khosla, Katz, K., Sidhaye, V.K., (2022). Epithelial plasticity in COPD results in cellular unjamming due to an increase in polymerized actin. **Journal of Cell Science**, 135(4).

16. Poudel, S., Chandrala, L., Das, D., and De, A. (2021). Characteristics of shock tube generated compressible vortex rings at very high shock Mach numbers. **Physics of Fluids**, 33(9), 96105.
17. Xue, X., Chandrala, L. D., and Katz, J. (2021). Flow structure and turbulence in the near field of an immiscible buoyant oil jet. **Physical Review Fluids**, 6(2), 24301.
18. Katre, P., Balusamy, S., Banerjee, S., Chandrala, L. D., and Sahu, K. C. (2021). Evaporation Dynamics of a Sessile Droplet of Binary Mixture Laden with Nanoparticles. **Langmuir**, 37(20), 6311–6321.
19. Afshar-Mohajer, N., Lam, A., Dora, L., Katz, J., Rule, A. M., and Koehler, K. (2020). Impact of dispersant on the crude oil content of airborne fine particulate matter emitted from seawater after an oil spill. **Chemosphere**, 256, 127063.
20. Ghosh, B., Reyes-Caballero, H., Akgün-Ölmez, S. G., Nishida, K., Chandrala, L., Smirnova, L., et al. (2020). Effect of sub-chronic exposure to cigarette smoke, electronic cigarette, and waterpipe on human lung epithelial barrier function. **BMC Pulmonary Medicine**, 20(1), 1–9.

Publications before joining IITH

21. Sampath, K., Afshar-Mohajer, N., Chandrala, L. D., Heo, W.-S., Gilbert, J., Austin, D., et al. (2019). Aerosolization of Crude Oil-Dispersant Slicks Due to Bubble Bursting. **Journal of Geophysical Research: Atmospheres**, 124(10), 5555–5578.
22. Chandrala, L. D., Afshar-Mohajer, N., Nishida, K., Ronzhes, Y., Sidhaye, V. K., Koehler, K., and Katz, J. (2019). A Device for measuring the in-situ response of Human Bronchial Epithelial Cells to airborne environmental agents. **Scientific Reports**, 9(1), 7263.
23. Murugan, T., Dora, C. L., De, S., and Das, D. (2018). A comparative three-dimensional study of impulsive flow emanating from a shock tube for shock Mach number 1.6. **Journal of Visualization**, 21(6), 921–934.
24. Kundu, A., De, S., Thangadurai, M., Dora, C. L., and Das, D. (2016). Numerical visualization of shock tube-generated vortex-wall interaction using a fifth-order upwind scheme. **Journal of Visualization**, 19(4), 667–678.
25. Dora, C. L., Murugan, T., De, S., and Das, D. (2014). Role of slipstream instability in formation of counter-rotating vortex rings ahead of a compressible vortex ring. **Journal of Fluid Mechanics**, 753, 29–48.
26. Murugan, T., De, S., Dora, C. L., Das, D., and Kumar, P. P. (2013). A study of the counter-rotating vortex rings interacting with the primary vortex ring in shock tube-generated flows. **Fluid Dynamics Research**, 45(2), 25506.
27. Murugan, T., De, S., Dora, C. L., and Das, D. (2012). Numerical simulation and PIV study of compressible vortex ring evolution. **Shock Waves**, 22(1), 69–83.
28. Ghosh, S. K., Dora, C. L., and Das, D. (2012). Unsteady wake characteristics of a flapping wing through 3D TR-PIV. **Journal of Aerospace Engineering**, 25(4), 547–558.
29. Dora, C. L., Saravanan, D., Karunakar, K., and Das, D. (2011). Characteristics of embedded-shock-free compressible vortex rings: a detailed study using PIV. **Advances in Mechanical Engineering**, 3, 650871.

* Corresponding author

PATENTS FILED

- Lakshmana D. Chandrala, K.K Bhardwaj, Debopam Das (2015), A system for particle generator in piv applications, Indian Patent, 4130/DEL/2015. Applicant: IIT Kanpur
- K. C. Sahu, Lakshmana D. Chandrala, P. S. Kolhe, P. K. Kirar, and S. S. Ade, A system to recreate dynamic weather conditions. Indian Patent No: 202341038384; Filing Date: 05/06/2023, Grant Date: 18/12/2023. Applicant: IIT Hyderabad
- Kishore Ramesh, Junaidvali Shaik, Lakshmana Dora Chandrala, Chandrika Prakash Vyasarayani, A Method and an Apparatus for Measuring Inertial Parameters of a Rigid Object, Indian Patent No: 202441040592. Applicant: IIT Hyderabad.
- Kishore Ramesh, Nizamuddin Khaderi Syed, Lakshmana Dora Chandrala, Chandrika Prakash Vyasarayani, An Apparatus to Measure a Load at a Fixed Joint, Indian Patent No: 202441041974. Applicant: IIT Hyderabad.
- Kishore Ramesh, Junaidvali Shaik, Lakshmana Dora Chandrala, Chandrika Prakash Vyasarayani, An Inertial Parameters Measurement System for a Rigid Object, Indian Patent No: 202441042132. Applicant: IIT Hyderabad.
- Gowri Padmaja Majji, Krishna Venkata Janam, Harish Nagaraj Dixit, Lakshmana Dora Chandrala, A System and Method for Measuring The Diameter of Multiple Micron Fibers, Indian Patent No: 202441085300, Granted: July 2025. Applicant: IIT Hyderabad.

- Gowri Padmaja Majji, Kirti Chandral Sahu, Lakshmana Dora Chandrala, An apparatus and a method to measure real-time rainfall rate based on digital holography, Indian Patent No: 202541076970, Filing Date: 13.08.2025. Applicant: IIT Hyderabad.

RESEARCH STUDENTS GUIDANCE

- PhD: Nine- (**Three graduated** + Four ongoing as main advisor + Two ongoing as co-advisor)
- As a Main Guide:
 1. V Vinod (MAE department, **PMRF**): Graduated 2025
 2. Someshwar S. Ade (ID program, **PMRF**): Graduated 2025
 3. Lokendra Mohan Sharma (MAE department, **PMRF**): ongoing
 4. Sai Krishnaprasad Gottipati (MAE department, External): ongoing
 5. Majji Gowripadmaja (MAE department): ongoing
 6. Shubham Chakraborty (ID program): ongoing
- As a Co-Guide:
 1. Charul Gupta (MAE department): Graduated 2024
 2. Kuruba Sai Akilesh (MAE department): ongoing
 3. Aman John Tudu (CHE department): ongoing
- M.Tech: Eight (six graduated + Two ongoing)

SPONSORED RESEARCH PROJECTS

S.No	Title	Cost (in Lakhs)	Duration	Role	Funding Agency
1	Development of shock tube facility to study blast mitigation	25.00	2020-2022	PI	IIT Hyderabad
1	Role of surfactants in break-up and dispersion of underwater oil jets: Application to oil-well blowouts.	33.15	2021-2023	PI	SERB (SRG)
1	Development of accurate rainfall models by studying the influence of meteorological parameters on raindrop dynamics on raindrop dynamics	97.50	2022-2024	Co-PI	SOCH, IIT Hyderabad
2	On the role of inertia and surfactants in moving contact line flows	56.85	2022-2025	Co-PI	SERB
3	Simulation of Aero-Optical Effects	9.90	2024-2025	PI	CHESS (DRDO)
4	Development of an unsteady wind tunnel for simulating flight-relevant flow conditions for a minor or nano UAV	270.12	2024-2028	Co-PI	DIA (DRDO)
5	High-dimensional Aerodynamic Data Modeling and Shape Optimization via Convolutional Neural Networks	19.17	2025-2026	PI	ASL (DRDO)
6	Modelling and establishment of various process parameters for PCS fibers and modelling of PCS-SiC conversion process	341.98	2025-2028	Co-PI	DIA (DRDO)
7	In situ measurements of aluminium agglomerate size, velocity, and temperature using simultaneous digital inline holography and imaging pyrometry	165.24	2025-2028	Co-PI	ARMREB (DRDO)

CONSULTANCY GRANTS

S.No	Title	Cost (in Lakhs)	Duration	Role	Funding Agency
1	Commercial vehicle cooling system layout concept evaluation for improving cooling performance	19.82	2024-2025	PI	Hyundai Motor India Engineering Pvt. Ltd
2	An analysis of the flow phenomenon in cooling grille holes	15.7	Jul-NOV 2023	Co-PI	Hyundai Motor India Engineering Pvt. Ltd

AWARDS AND RECOGNITIONS

- Three graduate students received the Prime Minister Research Fellowship (PMRF)
- Best oral presentation in 8th National Symposium on Shock Waves, 2024
- Best oral presentation in the International Conference on Experimental Mechanics, 2024.
- Johns Hopkins Postdoctoral Fellowship, 2016

ADMINISTRATIVE ACTIVITIES

- Faculty advisor: B.Tech 2021.
- PhD admission committee, 2021.
- Hostel Warden, 2021-2024.
- Faculty recruitment coordinator of the MAE department, 2021-2023.
- DPGC of the MAE department, 2025-present.

PROFESSIONAL ACTIVITIES

- *Reviewer*–Journal of Fluid Mechanics.
- *Reviewer*–American Institute of Aeronautics and Astronautics (AIAA).
- *Reviewer*–International Journal of Multiphase Flow.
- *Reviewer*–Industrial and engineering chemistry research.
- *Reviewer*–Powder Technology.
- *Reviewer*–Physics of Fluids.
- Member of American Physical Society (APS).
- Session Chair: Multiphase and particle-laden flows, 26th International Congress of Theoretical and Applied Mechanics (ICTAM 2024), South Korea.

SELECTED CONFERENCE PUBLICATIONS

1. Chandrala, L.Franz O'Meally, K., Katz, J. Effect of chemical herders on wave breaking. 72nd Annual Meeting of the APS Division of Fluid Dynamics. Bulletin of the American Physical Society. Seattle, USA, Nov 23-26, (2019)
2. Chandrala, L., Nishida, K., Afshar-Mohajer, N., Ronzhes, Y., Sidhaye, V., Koehler, K., et al. A novel exposure device to assess the health impacts of airborne particulate matter on the human respiratory system. In Gulf of Mexico Oil Spill and Ecosystem Science Conference 2019. New Orleans, LA , Feb 4-7, (2019).
3. Nishida, K., Chandrala, L., Afshar-Mohajer, N., Chen, S., Katz, J., Koehler, K., et al. in vitro Oily Marine Aerosol Exposure Alters Human Airway Epithelial Function. In Gulf of Mexico Oil Spill and Ecosystem Science Conference 2019. New Orleans, LA , Feb 4-7, (2019)

4. Afshar-Mohajer, N., Chandrala, L., Nishida, K., Sidhaye, V., et al, K. Development of an In Vitro Exposure System Enabled to Track Biological Impacts of Oily Aerosol on Human Lung Cells in Real-Time. In 10th International Aerosol Conference. Saint Louis, MO , Sep 2-7,(2018)
5. Chandrala, L., Afshar-Mohajer, N., Nishida, K., Sidhaye, R., Koehler, K., and Katz, J. A New In Vitro Exposure Device to Assess the Health Impacts of Oily Marine Aerosols on the Human Respiratory System. Gulf of Mexico Oil Spill and Ecosystem Science Conference. New Orleans, LA (2018)
6. Dora, C. L., Sampath, K., Afshar-Mohajer, N., Heo, W., Gilbert, J., Austin, D., et al. Aerosolization of crude oil-dispersant slicks due to popping of bubbles. *71st Annual Meeting of the APS Division of Fluid Dynamics*. Bulletin of the American Physical Society. Nov 18-20, (2018)
7. C.L.Dora, and Debopam Das., Velocity and density field of compressible vortex rings. *Aerospace engineering Doctorial student's Symposium-IIT Kanpur*, India, May, (2014)
8. D. Saravanan, C.L Dora, T. Murugan, S. Sankaran, Satyanarayana TNV and Debopam Das. Experimental investigation of flow and noise characteristics of impinging twin jets, simulating starting flow from a rocket booster on a launch pad. *STS-2011*, Thiruvananthapuram, Dec 16-18, 2011
9. S.K. Ghosh, C.L. Dora, G.K. Bunkar and D. Das, Understanding Flapping Flight through PIV Measurements on a Butterfly Simulating Model. *6th International Conference on Intelligent Unmanned Systems*, Bali, Indonesia, Nov 2-3, 2010
10. S.K. Ghosh, C.L. Dora and D. Das, Unsteady Wake characteristics of a flapping wing through 3D TR-PIV measurement, *6th International Conference on Intelligent Unmanned Systems*, Bali, Indonesia, Nov 2-3, 2010.
11. C.L.Dora, Murugan, T., Saravanan, S, Ghosh K. S., Sankaran S, Satyanarayana TNV and Das, D. Initial Acoustic load distribution on launch vehicle model during flow development. *3rd European conference for Aerospace Science- Paris*, France 6-9, July 2009
12. Debopam Das, Saurav K. Ghosh and C.L Dora. On Three-Dimensional Flow Field of Butterfly Flight. *Proceedings of the Indo-US Workshop on Micro Air Vehicles*, India, August 13-14 November 2008.